Ans. (c) : • The question can be answered by using both the statements together, but cannot be answered using either statement alone. There is only one situation, which satisfied the condition–

if $m = 24$

$n = \perp$

Statement $\perp$: $m + n > mn$ and $m > n$

$24 + 1 > 24$ and $24 > 1'$

Statement 2 : $m \times n = 24$

$24 \times 1 = 24$

By using both statements we can give the answer.

70. A Question is given followed by two Statement I and II. Consider the Question and the Statements.
Question: What is the time required to download the software?
Statement–I: The size of the software is 12 megabytes.
Statement–II: The transfer rate is 2.4 kilobytes per second.
Which one of the following is correct in respect of the above Question and the Statements?
(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.
(b) The Question can be answered by using either Statement alone.
(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.
(d) The Question cannot be answered even by using both the Statements together.

Ans. (c) : Time required to download the software

$$(T) = \frac{\text{Size of software}}{\text{Transfer rate}}$$

To calculate time of downloading we require both size and rate of downloading.
Hence the question can be answered by using both the statements together, but cannot be answered using other statement alone.

Directions for the following 4 (four) items:
Read the following two passages and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage–1

Conventional classrooms, by emphasizing fixed duration over learning effectiveness, resign themselves to variable outcomes. The tyranny of the classroom is that every learner is subjected to the same set of lectures in the same way for the same duration. In the end, a few learners shine, some survive, and the rest are left behind. After the fixed duration, the classroom model moves on, with not a thought spared for those left behind. This is how we end up with 10 percent employability in our graduates after a decade and half of formal education. Repeating the same ineffectual script in the realm of skill education will not produce different results.

71. Which of the following statements best reflects/reflect the most logical and rational inference/inferences that can be made from the passage?
1. In conventional classroom learning, the central goal is duration of learning rather than attainment of competency.
2. Conventional classrooms encourage one-size-fits-all approach and stamp out all differentiation.
Select the correct answer using the code given below:
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : Inference 1 is correct, because the statement" in conventional classroom learning the central goal is duration of learning rather than attainment of competency, reflect according to the passage. In passage discusses all approach in the form of fixed class durations and the same set of futures for all.
• **Inference 2 is also correct,** because, in the passage it is defined that a conventional classrooms encourage one size fits all approach and stamp out all differentiation. In conventional classrooms a learner emphasizes and fixed duration over learning effectiveness.
• After the fixed duration, the classroom model moves on, with not a thought spread for those left behind.

72. Based on the above passage, the following assumptions have been made:
1. As a large number of workers in our country are employed in unorganized sector, India does not need to change its present conventional classroom system of education.
2. Even with its present conventional classroom system of education, India produces sufficient number of skilled workers to fully realize the benefits of demographic dividend.
Which of the assumption given above is/are valid?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : • **Inference 1 is not correct,** because the statement "As a large number of workers in our country are employed in unorganized sector, India does not need to change its present conventional classroom system of education" does not define the perfect theme of the passage. The passage defines the conventional classroom, by emphasizing fixed duration over learning effectiveness, resign themselves to variable outcomes.
• **Inference 2 is not correct,** because the passage discusses the issues with conventional education system and linking them to poor employability or lack of realization of our demographic dividend. So, the second assumption is not appropriate according to the passage.

Passage–2

When a child reaches adolescence, there is apt to be a conflict between the parents and the child, since the latter considers himself to be by now quite capable of managing his own affairs, while the former are filled with parental solicitude, which is often a disguise for love of power. Parents consider, usually, that the various moral problems which arise in adolescence are peculiarly their province. The options they express, however, are so dogmatic that the young seldom confide in them, and usually go their own way in secret.

73. Based on the above passage, the following assumptions have been made:
1. The adolescent does not feel comfortable with his parents because they tend to be dominating and assertive.
2. The adolescent of modern times does not have much respect for parents.
Which of the assumptions given above is/are valid?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : • **Assumption 1 is valid,** because the statement" the adolescent does not feel comfortable with his parents, because they tend to be dominating and assertive is appropriate according to the passage. when a child reaches adolescence, the opinions of parents and child gets change. It clarifies that parents are dominating and assertive, making adolescents uncomfortable.
• **Assumption 2 is not valid,** The passage discusses the behaviour and dynamics between adolescents and parents. It highlights lack of confiding of adolescents in their parents due to the latter's dogmatic behavour, This way we find that the second assumption is not appropriate according to the passage.

74. Which one of the following statements best reflects the central idea of the above passage?
(a) Parents in general may not be of much help when children are on their way to becoming adults.
(b) When children reach adolescence, involvement of parents in their lives is unnecessary.
(c) Modern-day nuclear families are not capable of bringing up children properly.
(d) In modern societies, adolescents tend to be stubborn, disobedient and careless.

Ans. (a) : Options A is correct- The central idea of the passage that adolescents often perceive their parents as dominating and assertive, leading them to ovoid confiding in their parents and instead navigate their own path independently. The passage discusses how parental appeals can be seen as a disguise for authority, causing teens to keep their opinions to themselves.
• **Options B is in correct,** The passage criticise unnecessary involvement but does not suggest complete lack of need for parental involvement.
• **Options C is in correct,** The passage quite general and makes on special type of family structure where this pattern is observed.
• **Options D is in correct,** The passage focuses more on the perception of parental dominance and adolescents secretive behaviour instead of general characteristics of adolescence.

75. What is the number of fives used in numbering a 260-page book?
(a) 55 (b) 56 (c) 57 (d) 60

Ans. (b) : Number of fines used in 1 to 100, is
5, 15, 25, 35 95
at unit place no. of 5 = 10 times
10th place no. of 5 = 10 times
Between 101 to 200 –
at unit place, no. of 5 = 10 times
10th place no. of 5 = 10 times
Between 201 to 260 –
at unit place no. of 5 = 6 times
10th place no. of 5 = 10 times
So, total. no. of five used 56 times.

76. What is the sum of the first 28 terms in the following sequence?
1, 1, 2, 1, 3, 2, 1, 4, 3, 2, 1, 5, 4, 3, 2,.....
(a) 83 (b) 84 (c) 85 (d) 86

Ans. (b) : First 28 terms in the following sequence–
1 1 2 1 3 2 1 4 3 2 1 5 4 3 2 1 6 5 4 3 2 1 7 6 5 4 3 2 1
Sum = 2 + 3 + 6 + 10 + 15 + 21 + 27 = 84

77. A person buys three articles, P, Q and R for ₹ 3,330. If P costs 25% more than R and R costs 20% more than Q, then what is the cost of P?
(a) ₹ 1,000 (b) ₹ 1,200
(c) ₹ 1,250 (d) ₹1,350

Ans. (d) : According to question ratio will be–

	P	:	Q	:	R
Cost price →	15	:	10	:	12

Cost price of P $= \frac{15}{(15+12+10)} \times 3330 = 1350$ ₹

78. If the sum of two-digit numbers AB and CD is the three-digit number 1CE, where the letters A, B, C, D, E denote distinct digits, then what is the value of A?
(a) 9 (b) 8
(c) 7
(d) Cannot be determined due to insufficient data

Ans. (a) : According to given condition in the question–

$$\begin{array}{r} A\ B \\ +\ C\ D \\ \hline 1\ C\ E \end{array}$$

if B + D = E and 1 is carry for word then–

$$\begin{array}{r} 1 \\ A\ B \\ +\ C\ E \\ \hline 1\ C\ E \end{array}$$

$\Rightarrow$ $1 + A + C = 1C$
in this situation only 9 at the place of A would be correct digit that will give C after adding C.

79. Three numbers x, y, z are selected from the set of the first seven natural numbers such that x > 2y > 3z. How many such distinct triplets (x, y, z) are possible?
(a) One triplet (b) Two triplets
(c) Three triplets (d) Four triplets

Ans. (d) : Set of first seven natural numbers are 1, 2, 3, 4, 5, 6,7
$\Rightarrow$ x > 2y > 3z, x, y & z are distinct value.
x > 2y > 3z
1st possibility → x = 7, z = 1, y = 2 $\Rightarrow$ 7 > 4 > 3
2nd possibility → x = 7, y = 3, z = 1 $\Rightarrow$ 7 > 6 > 3
3rd possibility → x = 6, y = 2, z = 1 $\Rightarrow$ 6 > 4 > 3
4th possibility → x = 5, y = 2, z = 1 $\Rightarrow$ 5 > 4 > 3
So, there are total 4 such distinct triplets are possible.

80. The total cost of 4 oranges, 6 mangoes and 8 apples is equal to twice the total cost of 1 orange, 2 mangoes and 5 apples. Consider the following statements:
1. The total cost of 3 oranges, 5 mangoes and 9 apples is equal to the total cost of 4 oranges, 6 mangoes and 8 apples.
2. The total cost of one orange and one mango is equal to the cost of the one apple.
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : Given–
4 Orange + 6 Mangoes + 8 Apples = 2 (1 Orange + 2 Mangoes + 5 Apples)
After rearranging –
1 Orange + 1 Mango = 1 Apple
Statement 1: –
3 Orange + 5 Mangoes + 9 Apple = 4 orange + 6 Mangoes + 8 Apple
After rearranging– 10 oranges + 1 Mangoes = 1 Apple
So, Statement 1 is correct.
Statement 2 :–
1 Orange + 1 Mango = 1 Apple
Statement 2 is also correct statement